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Editor-in-Chief
Education Sciences
MDPI

Dear Editor,

We are pleased to submit our manuscript entitled “Deep Learning for Student Engagement Detection: A Lightweight CNN Framework with Channel-Spatial Attention for Educational Environments” for consideration for publication in *Education Sciences*.

Automated student engagement detection is a critical challenge in modern educational environments, particularly with the rapid expansion of online and blended learning. Our paper proposes a lightweight deep learning framework that integrates MobileNetV3 with a novel Channel-Spatial Attention Module (CSAM) for real-time engagement recognition from facial expressions. The key contributions of this work are:

1. **A lightweight CSAM module** specifically designed for mobile-class architectures, achieving 1.3–5.0% accuracy improvement with only 0.01M additional parameters.
2. **A comprehensive evaluation across three datasets:** FER2013 (70.5% accuracy, 5-fold CV), CK+ cross-dataset evaluation (71.9% without fine-tuning), and DAiSEE engagement-specific validation (61.2% test accuracy on binary classification).
3. **A principled emotion-to-engagement mapping** grounded in Fredricks’ multidimensional engagement theory, validated through per-class analysis and cross-dataset generalization.
4. **Deployment-oriented evaluation** including parameter count, FLOPs, inference latency, and energy consumption, demonstrating that competitive accuracy does not require heavy architectures.

This work is particularly relevant to *Education Sciences* as it addresses the practical need for efficient, deployable engagement monitoring systems in real educational settings. The MobileNetV3+CSAM model achieves 70.5% accuracy with only 1.26M parameters and 53 fps inference speed, making it suitable for edge deployment in classrooms.

All authors have read and approved the manuscript. The work has not been published elsewhere and is not under consideration by another journal. The authors declare no conflicts of interest.

We believe this manuscript makes a meaningful contribution to the field of educational technology and would be of interest to the readership of *Education Sciences*. We look forward to your decision.

Sincerely,

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